

Tripartite Space Cybersecurity Resilience Model: Computed Results, Figures, and Tables

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Generated from live model output

All values below are live output from the reference implementation (calibrated parameter set; finite-adversary cap active; $n = 63$ disclosed incident corpus; 10-fold cross-validation with seed 42). Figures are vector PDFs produced by the analysis pipeline. Tables are emitted by `make_tables.py` and reproduce the paper’s Table V–VII family with recomputed numbers.

1 Result Tables

Table 1: Baseline Resilience Trajectory (Model-Conditional)

Year	R_{sys}	R_{eff}	Threat	Erosion
0	0.283	0.283	0.100	0.0%
3	0.635	0.599	0.248	5.8%
5	0.726	0.658	0.343	9.5%
8	0.810	0.699	0.451	13.7%
10	0.864	0.728	0.503	15.7%
15	0.917	0.746	0.580	18.6%
20	0.928	0.744	0.609	19.8%

Table 2: Comparative Performance Against Baseline Models (10-fold CV, disclosed $n = 63$)

Model	RMSE	r	MAPE
Naive mean	0.180	-0.44	34.2%
Static risk (FAIR)	0.373	0.86	64.9%
Resilience triangle	0.142	0.99	24.6%
Linear additive	0.076	0.90	12.3%
Independent dimensions	0.101	0.82	11.9%
Proposed (full)	0.064	0.93	10.0%

Table 3: Strategy Performance at Year 10 (Model-Conditional)

Strategy	R_{sys} (yr10)	R_{eff} (yr10)	Δ vs base
Baseline	0.864	0.728	–
Preparedness	0.877	0.739	+1.4%
Resistance	0.828	0.701	-3.7%
Restoration	0.874	0.736	+1.0%
Adaptation	0.867	0.731	+0.4%
Supply Chain	0.874	0.736	+1.0%
Holistic	0.932	0.777	+6.7%

Table 4: Parameter Sensitivity (Sobol Indices, Year 10)

#	Parameter	S_i	S_{Ti}
1	γ_{threat}	0.489	0.504
2	θ	0.269	0.272
3	$k_{restore}$	0.081	0.081
4	δ_{threat}	0.056	0.058
5	k_{supply}	0.031	0.032
6	β	0.024	0.023
7	ϕ	0.015	0.012
8	k_{prep}	0.009	0.012

Table 5: Operational-Constraint Scenario Effects (C1–C3)

Constraint	Metric	Value
Reference	Holistic advantage (yr10)	6.7%
C1 budget	Holistic advantage (yr10)	7.6%
C2 cap (0.85)	Erosion (yr20)	17.9%
C2 cap (0.85)	Tax mult. yr20 (capped)	1.22×
C2 uncapped	Tax mult. yr20	1.54×
C3 ($T_{cyc}=5$)	Mean resilience gain	+0.3%
C3 ($T_{cyc}=5$)	Cumulative cost	+22.0%

Table 6: Year-10 Policy Conclusions under Adverse Prior Perturbation

Conclusion	Baseline	Adverse Shift
Holistic advantage	6.7%	6.7%
Supply-chain leverage L	1.71	1.71
Red Queen erosion	15.7%	18.2%

Table 7: Retrodictive Case-Study Posture Indices

Case	AHP Posture	Observed Outcome
Viasat KA-SAT (2022)	0.272	Catastrophic; $\approx$ 45-day physical recovery
Starlink (2022)	0.472	Localized; days-scale disruption

Table 8: Calibrated Parameter Set with Uncertainty and Identifiability Class

Parameter	Sym.	Value	80% CI	Class
Preparedness decay	k_{prep}	0.05	[0.035, 0.065]	Practical
Resistance decay	k_{resist}	0.07	[0.049, 0.091]	Practical
Restoration decay	$k_{restore}$	0.12	[0.084, 0.156]	Structural
Adaptation decay	k_{adapt}	0.03	[0.021, 0.039]	Structural
Supply chain decay	k_{supply}	0.11	[0.077, 0.143]	Practical
Latency sensitivity	α	0.85	[0.6, 1.11]	Practical
Inaccessibility	β	1.2	[0.84, 1.56]	Practical
Radiation sensitivity	γ	0.9	[0.63, 1.17]	Practical
Thermal sensitivity	γ_{T}	0.15	[0.1, 0.2]	Weak
Superlinear exponent	μ	0.8	[0.56, 1.04]	Structural
Adversary learning	θ	0.12	[0.084, 0.156]	Practical
Adversary amplification	ϕ	0.25	[0.17, 0.33]	Structural
Threat decay	δ_{threat}	0.08	[0.056, 0.104]	Structural
Max threat impact	γ_{threat}	0.35	[0.25, 0.46]	Structural
Min threat capability	S_{threat_min}	0.1	[0.07, 0.13]	Practical
SC risk sensitivity	ϵ	0.45	[0.32, 0.58]	Weak
Complexity sensitivity	ν	0.3	[0.21, 0.39]	Weak

2 Figures

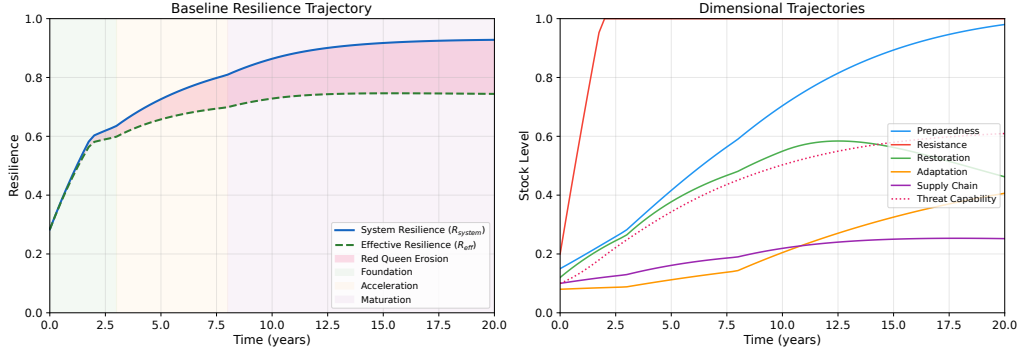


Figure 1: Baseline resilience trajectory with credible intervals; three growth phases (foundation, acceleration, maturation) annotated.

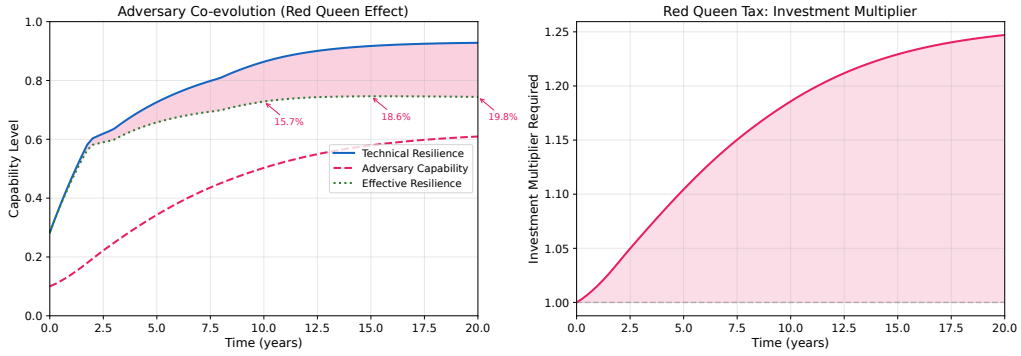


Figure 2: Technical vs. effective resilience under adversary co-evolution; Red Queen erosion shaded, annotated at years 10 and 20.

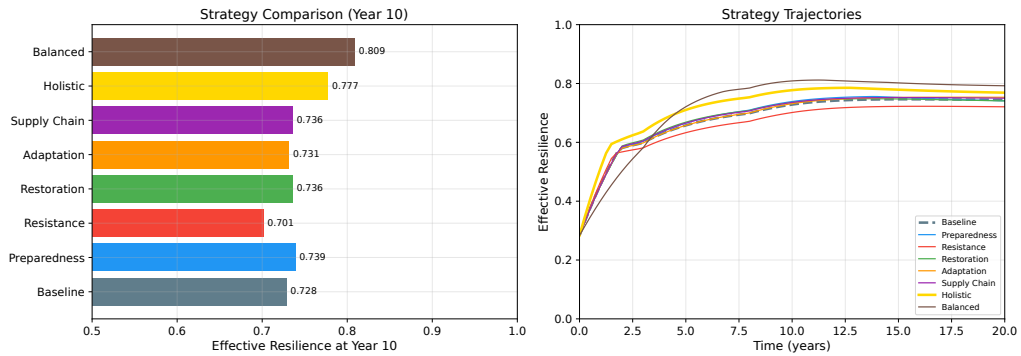


Figure 3: Strategy comparison: holistic vs. specialized single-dimension allocations over the mission lifecycle.

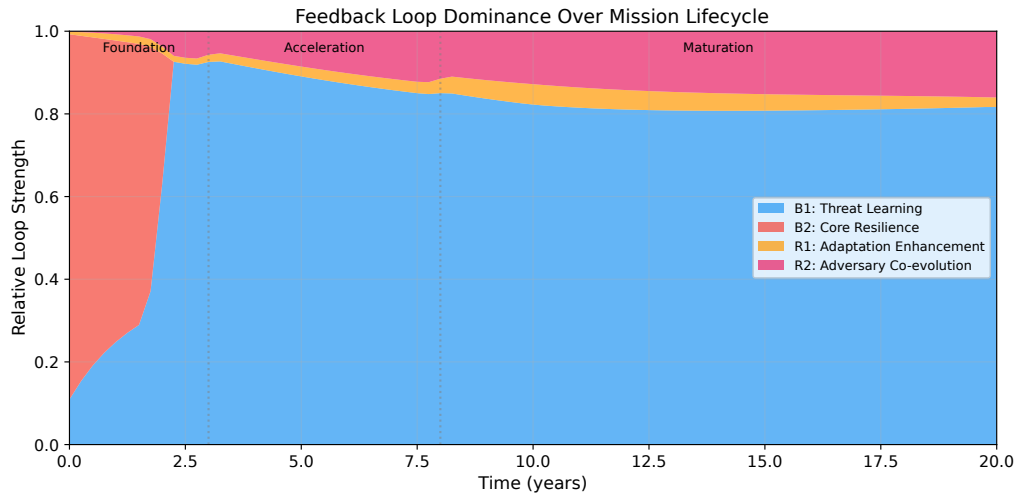


Figure 4: Feedback-loop dominance (B1, B2, R1, R2) over the mission lifecycle, producing the Red Queen plateau.

Monte Carlo Output Distributions

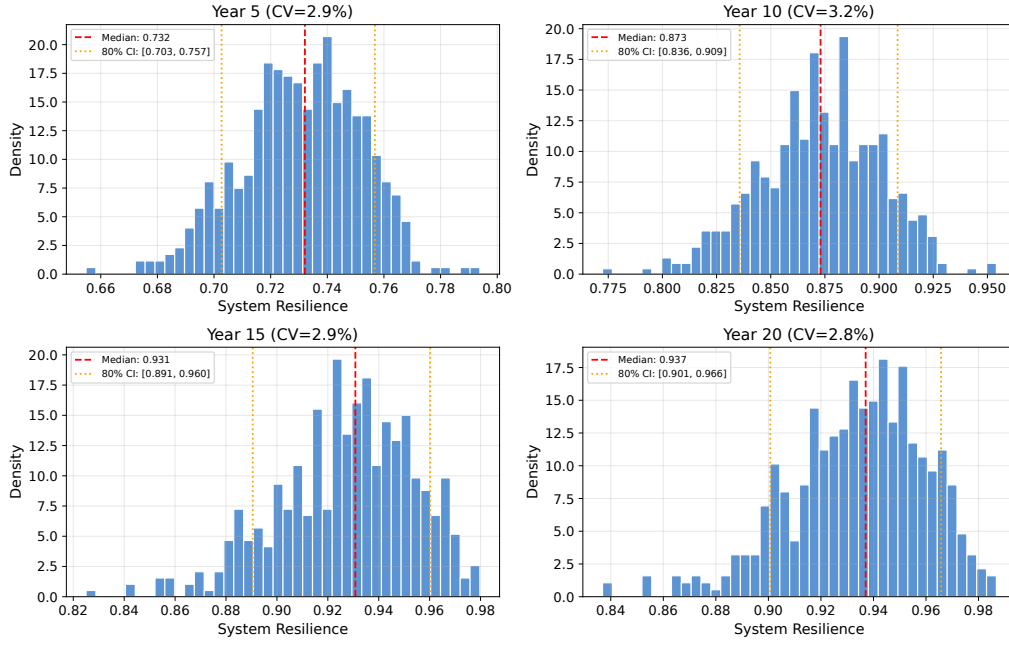


Figure 5: Monte Carlo output distributions for system resilience at years 5, 10, 15, and 20.

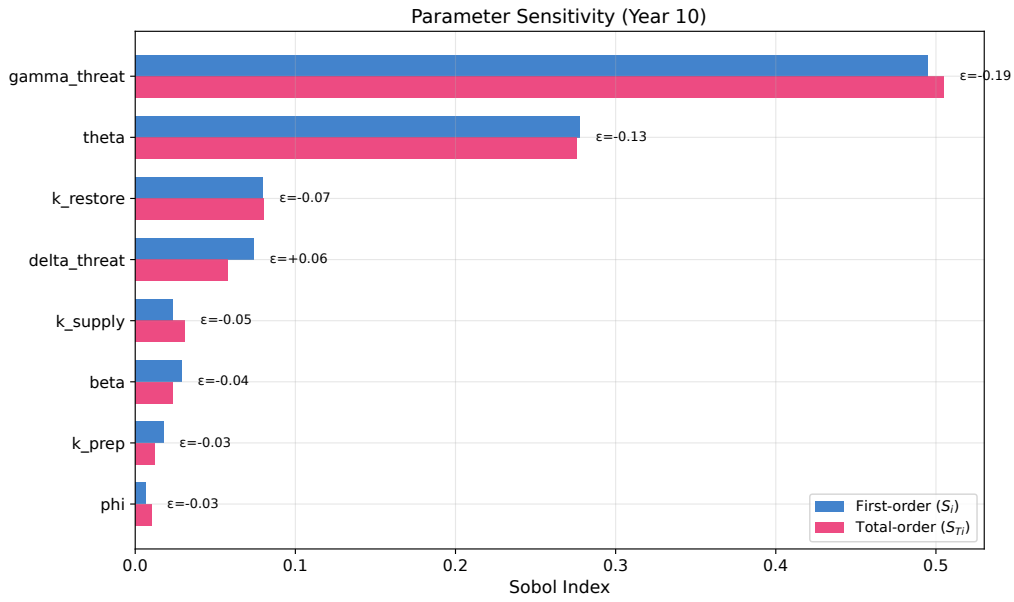


Figure 6: Parameter sensitivity: first- and total-order Sobol indices at year 10.

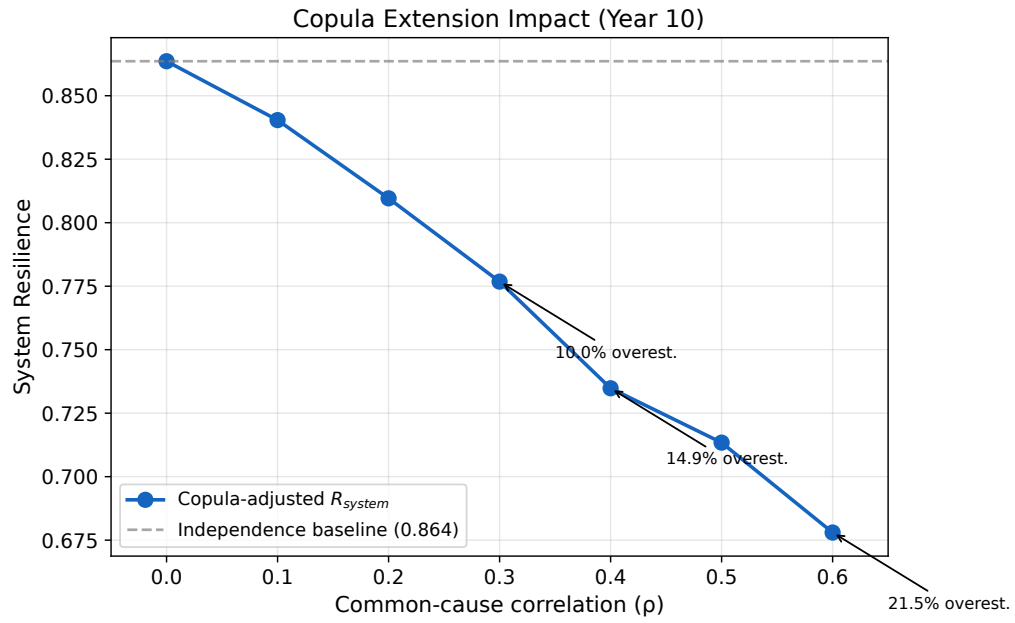


Figure 7: Copula-based correlated-failure analysis across $\rho \in [0, 0.6]$.

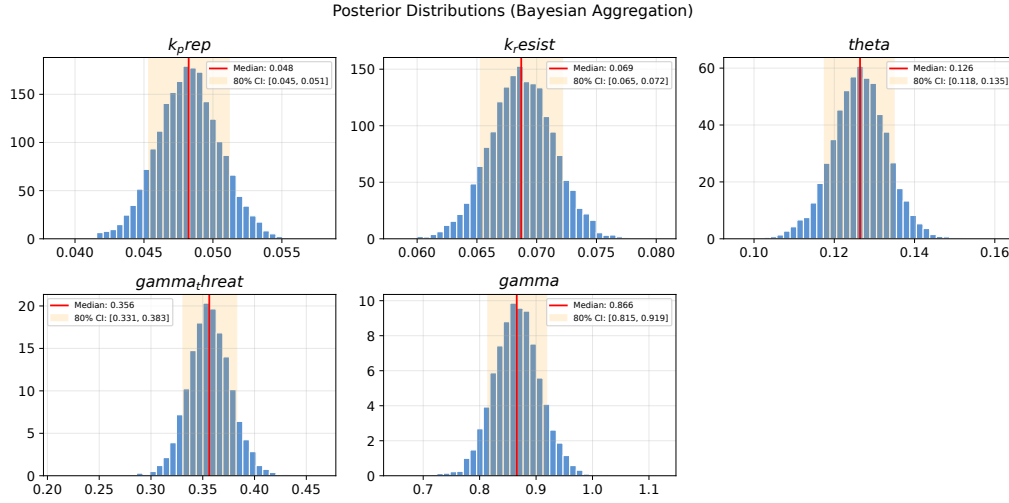


Figure 8: Posterior distributions of six key parameters from Bayesian aggregation of expert judgments.

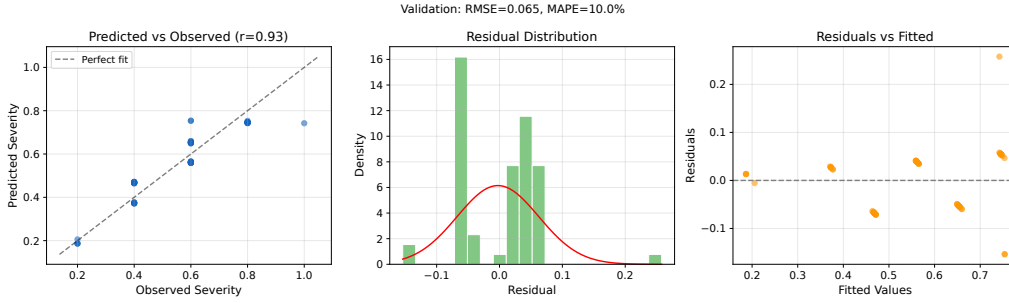


Figure 9: Residual analysis for historical validation on the disclosed corpus.